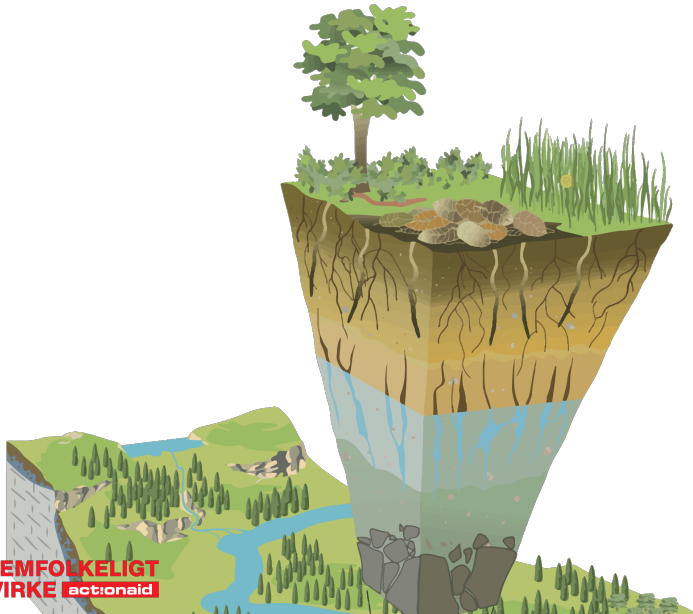


Don't Shit where You Drill!



Don't Shit where You Drill!

Michel Hardenberg

PhD student at AU - Quantifying the Vadose Zone
Department of Electrical and Computer Engineering

**MELLEMFOLKELIGT
SAMVIRKE **

CLIMATE JUSTICE SOCIETY JUNE 13, 2026

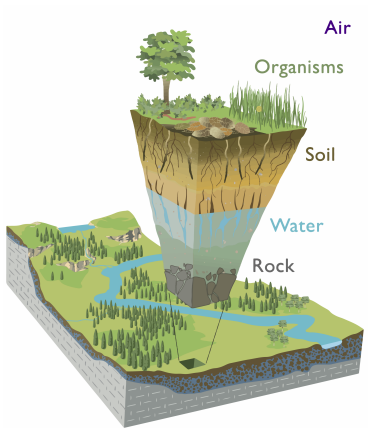


Illustration of soil layers in the near surface [1].

ToC

Part One:

- ▶ What are the culprits?
- ▶ How are we looking?
- ▶ What is the government doing? Promising?

Nitrate - the Classic

' For over 60 years, the global guideline for nitrates in drinking water has been 50 milligrams per litre (mg/L) [...] set in the 1950s¹.'

This limit is hopelessly out of date

INCREASED RISK OF COLORECTAL CANCER

“Nitrate in drinking water is a contaminant which can affect human health and has been associated with an increased risk of, amongst other diseases, colorectal cancer. [...] The analysis shows that a lower limit of 9.25 mg/L would decrease the annual number of colorectal cancer cases by 72 [...] and by an additional 55 % [...] for a stricter limit of 3.87 mg/L.” [2] [..?? different ref]

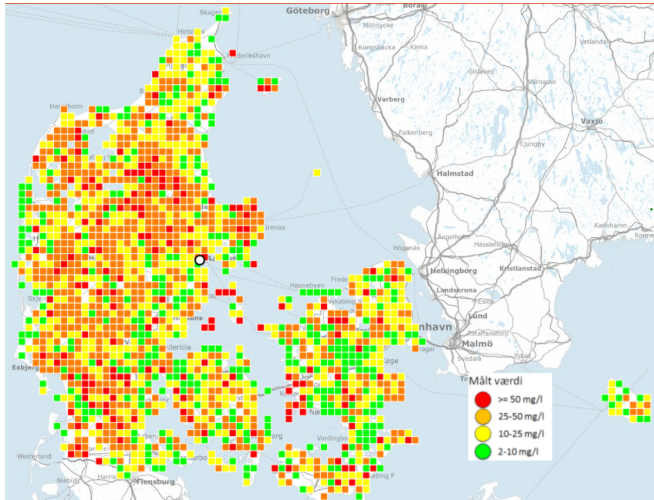
BLUE BABIES (NOT GOOD)



Infant methemoglobinemia

- ▶ Nitrate prevents oxygen from binding to their blood. Can progress to cause coma and death if not left untreated [3, 4].
- ▶ Water used in formula contained nitrate-nitrogen concentrations of 22.9 and 27.4 mg/L [3].

NITRATE POLLUTION IN DENMARK



<https://www.geus.dk/produkter-ydelser-og-faciliteter/data-og-kort/grundvandskort-og-data>

MIXING AND EXPLOTING

PFAS - the Immortal

WHERE DO THEY LIVE?

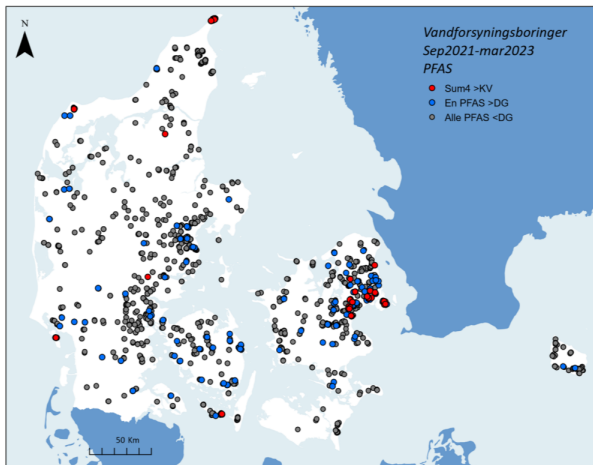
Where **DON'T** they live!? PFAS are found in water, air, fish, and soil..

WHERE DO THEY LIVE?

Where DON'T they live!? PFAS are found in water, air, fish, and soil..

Also, in blood of people, animals, in food products²

PFAS MEASUREMENTS IN GROUNDWATER WELLS



Red: PFAS chemicals above limit. *Blue:* Measured PFAS chemicals. *Grey:* None detected [5].

ToC

Part Two:

- ▶ Case studies in the global south
- ▶ How are we as Europeans involved?
- ▶ How is climate change changing the picture?

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